

Who Gets to Learn Data? A Co-design Framework for Popular Data Literacy Education in Underserved Communities

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Fig. 1. Illustrated panorama of a hillside urban community in Brazil, representing the socio-technical context in which community data practices and data literacy emerge.

Brazilian guidelines and curricula emphasize the importance of fostering data literacy in basic education; however, there is a gap in the provision of basic data knowledge, especially for residents of socially vulnerable communities, who have historically faced difficulties in having their demands met by public authorities. This article uses the epistemological-methodological approach of design science research to support the design of a framework aimed at helping teachers and instructional designers in creating strategies to foster data literacy among residents of socially vulnerable communities, starting from the results of an ethnographic study focused on understanding aspects of data interaction among residents of the Complexo do Alemão community in Rio de Janeiro. As a result, we discovered four dimensions of a framework that is informed by the DSR relevance cycle and will be informed by other ethnographic findings in dialog with Socio-cultural-historical Psychology meeting Popular Education approach.

CCS Concepts: • **Human-centered computing** → **Visualization design and evaluation methods**.

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1 Introduction + Position

Data literacy has been necessary for some time now, as data is everywhere and data-driven technologies enable everything — from everyday tasks like streaming, content optimization, and personalized learning to less common activities such as virtual property transactions in the metaverse and natural resource management [4]. Despite the launch of the Brazilian Computing curriculum for basic education in 2022 [1], there is a significant limitation in current approaches to data literacy in peripheral regions, especially regarding popular education approaches. Therefore, there is a need to create means for data literacy connected with the community's problems and virtues, addressing collaborative and civic aspects for active participation in public life and political decisions that affect them.

Given the need to design data literacy strategies situated in territorial contexts, this study draws on prior work that proposes frameworks for educational data strategies. Moon et al. [9], building on the structure proposed by Ridsdale et al. [12], developed a conceptual framework through three stages: a review of data literacy concepts and components in elementary education, an analysis of the Korean curriculum to identify data literacy elements across subjects and schooling levels, and a validation process based on expert interviews. Olari and Romeike [10] (2021) approached data literacy from the perspective of artificial intelligence education, showing that existing AI literacy frameworks insufficiently address core data literacy competencies across the data lifecycle; in response, they proposed a framework that integrates AI-related skills more holistically. More recently, Brito [2] proposed a framework for designing pedagogical data visualization workshops in high school, grounded in situated practices and problem-posing education. Her approach structures workshops around socially relevant questions and collective reflection, using Charles Maguerez's arc to support hypothesis formulation and intervention design.

Embodied interaction concerns how meaning is created, established, and communicated through the incorporation of technologies into practice [6]. This thinking applies directly to research on human interaction with data in vulnerable communities for the design of learning strategies, in search of meaning in the use of data constructed socio-historically, emerging from the culture and daily practices of workers [7] [8] through the performance of collaborative activities with the use of artifacts [13]. Our theoretical assumption is that data literacy learning experiences for children will be more meaningful when they integrate historical perspectives on data, civic engagement practices, and collective meaning-making, rather than focusing solely on technical skills. The research question is: **how can data literacy be understood and transformed into meaningful pedagogical practices in socially vulnerable communities?**

Design entails a conscious effort to impose meaningful order [11]. However, in this research, such "order" does not emerge from top-down prescriptions but from ethnographic engagement with situated data practices, where meaning is collectively produced and negotiated. Considering that frameworks need to emerge from situated research practices, this research aims to collaboratively develop a framework to support pedagogical teams and instructional designers in creating educational resources that foster data visualization literacy in socially vulnerable communities. The framework will be developed based on extensive ethnographic research already conducted with the communities of Complexo do Alemão and Complexo da Penha in Rio de Janeiro.

1.1 Methodological grounding

This research comprised Design Science Research epistemological methodological approach with ethnographic work over two years of interaction with the communities of Complexo do Alemão and Complexo da Penha. Both communities are located in territories controlled by parallel power structures, characterized by low public investment in infrastructure, symbolic and state violence through police incursions, and a high concentration of workers and children of native Brazilian origin and from victims of transatlantic trafficking for forced labor. The ethnographic work in this research is participatory and critical-interpretive, inspired by the Paulo Freire Method [8] and Asset-Based Design [14] [5][3], considering the need to construct knowledge in a situated manner, taking into account power relations, inequalities, and political implications related to the interaction of vulnerable people with data. The researcher's role was that of an external person to the field, yet close due to sharing a similar cultural, racial, and territorial origin with the residents. The researcher moved between observer, participant observer, and mediator in the residents' interaction with data.

This research was designed to conduct three cycles of DSR, in which medium artifacts are created and applied as design objects to investigate human interaction with data. Artifacts evolve through three DSR cycles – the relevance cycle (first cycle), the design cycle (second cycle), and the rigor cycle (third cycle) – toward the design of the research output artifact. The artifact output will materialize into a framework to help teachers foster data literacy among people living in socially vulnerable areas.

The relevance cycle is designed to give visibility to data concepts and practices of the participating communities, technological limitations that occur locally for learning in these contexts, as well as the demand for situated and embodied practices. The relevance cycle utilizes artifacts constructed with the intention of understanding the participants' actual level of development. At the end of the relevance cycle, the research problem is validated. Following the relevance cycle, the instrumental artifacts evolve technologically and psychologically to compose the material of the rigor cycle.

The design cycle expands practices already partially mastered by the community, placing the local knowledge discovered in the first cycle into dialogue with Vygotsky's socio-historical psychology towards the design of more refined strategies for data literacy and adjustment of the already emerging framework through data collected in the interactions. Following the design cycle, the instrumental artifacts evolve technologically and psychologically to compose the material for the rigour cycle. At the end of the design cycle, the artifact means and the artifact outcomes are evaluated.

In the rigour cycle, the artifacts evolve to demand greater abstraction and autonomy from the participants in working with data. The rigour cycle is built to verify whether the framework proposal presented in the design cycle supports real developmental progress for participants towards data literacy. In this cycle, the role of human mediation will be adjusted, reduced, and redistributed so that the learner approaches the actual level of development through the internalization of processes. At the end of the rigour cycle, it is expected to obtain the consolidated framework plus a set of exemplary artifacts at various levels to compose educational strategies. Is also expected to conduct a theoretical-analytical evaluation of the results, including a discussion of validity, comparison with the literature, and identification of limitations.

2 Generative ethnographic insights

To inform the design of the framework, we collected data through i. semi-structured interviews conducted with the pedagogical team working on a municipal social project focused on promoting community interaction with technology; ii. participant observation in group classes held with university students, professors, community leaders, representatives

from the Rio de Janeiro city government's science and technology department, and guests; iii. During the relevance DSR cycle, data literacy workshops were held with the communities to understand how residents interact with data using simple artifacts and understand situations that arise from this cycle of interaction. The use of unplugged artifacts is intentionally restricted to the initial design cycle, serving as a baseline level of mediation from which more complex artifacts will progressively be introduced. Some ethnographic insights that inform the framework are:

Insight 1: Residents of territories with little or no government assistance need to learn about data because it provides social emancipation, highlighting the characteristics (problems and virtues) of the communities, and supporting the struggles organized by the Third Sector. In the participants' statements:

"Perhaps there is a process of persuasion that precedes the learning process. A process of scientific dissemination, of presenting data as an opportunity and transformation of lives [...] There is a lot of utilitarianism in most of these communities, so people will grow up and become soldiers, or sometimes motorcycle taxi drivers, barbers, beauticians - very dignified professions that carry many traditions and cultures important to our territories, but which sometimes also hide the lack of opportunities, the lack of access to new horizons, perceptions of value, learning and computational skills." (Participant 2)

Design should consider scientific dissemination as preceding the task of formal education, so that people, upon having their first contact with the topic of data literacy, can develop an interest in the area and realize they are capable of being protagonists in their relationship with data, both for personal enjoyment and in the world of work, and in the possibility of using it as a means to express the problems and virtues of their community.

Insight 2: The community has few educational resources to foster data literacy among its users. And the technological resources, when used, are very underutilized.

"At the Nova Brasília Knowledge Center, we really got into the data area after you joined. Because until then, we didn't have it. I believe we didn't. Back when [employee's name] was there, it wasn't that important, you know. We didn't know. We also didn't understand how to incorporate it into the daily lives of these students. So that kind of turned the key after you arrived. Obviously, the teachers in the technology area [...], they work with it much more than we do. And there's Excel too. Excel uses a lot of [data]." (Participant 9)

The design should consider utilizing the technological resources that the Knowledge Center possesses, in addition to adding possibilities for using other technologies to build playful and situated experiences, directly engaging with the reality of the residents, taking care not to fall into technicality, and focusing the use of tools on the needs and contradictions perceived by the residents in their daily lives.

Insight 3: Young adults in the community have a limited educational background in the technical aspects of data literacy, but they understand important parts of the complexities and contradictions related to the new data paradigm in capitalist society.

“But I think what happens there [in the episode] also happens today. It’s how much it intensifies the negative. Because, for example, she received more than fifteen five-star ratings. Then she received a four-star rating. Then she stopped and looked at whose profile it was and became reflective. And that happens today, right? And hate is what generates a lot of engagement on the internet. So, I think technology makes us focus on what is negative and bad, right? It messes with emotions, but mostly in that way. We sometimes ignore the good side and focus on what’s bad, on what generates hate in us.” (Participant 29)

Design should consider utilizing the level of awareness that residents already have regarding the social issues involved in data use to promote data literacy learning, incorporating themes and discussions into activities that also aim to increase their level of social awareness.

3 From ethnographic insights to an instructional design framework

A framework is a working structure, a kind of blueprint, from which something can be created. Its main role is to provide a basic guideline scheme to save time and effort in the development of products or services [2]. The next steps in this research involve abstracting the insights obtained through ethnographic work, grouping the insights according to emerging categories, deciding whether it will be necessary to use any conceptual or curricular basis to better define the framework’s categories, and whether it will be necessary to enrich the set of categories obtained with information from other pre-existing structures. Up to this point, we consider that there are four categories that have stood out:

Visualization literacy: This refers to the representational dimension of the framework, which explores ways to make data perceptible and interpretable, including graphical representations and material externalizations, as well as transitions between the concrete and the abstract. The theoretical basis for the development of this dimension lies in human-data interaction, data visualization, and Vygotsky’s theory of development.

Interaction: This refers to the normative and reflective dimension of the framework, which addresses social and ethical awareness in the use of data. It should include dimensions such as human positioning in relation to presented narratives, critical reading, and positioning in relation to data visualizations, as well as the ability to collaborate with people and machines to build knowledge through data. The theoretical basis for the development of this dimension lies in Vygotsky’s zone of proximal development, computer-supported collaborative work, and human-data interaction.

Development: This refers to the sociocultural and contextual dimension of skills development. It should consider the development of competencies over time, as well as the training of critical thinking in real-world contexts, and should be geared towards solving problems of interest to the community. The theoretical basis for the development of this dimension lies in sociocultural psychology and popular education.

Agency & Advocacy: This refers to the dimension of awareness that leads to informed action. It should consider the use of data to intervene in the world, advocating for community interests, working towards informed decision-making, and making claims through data. The theoretical basis for the development of this dimension lies in critical pedagogy, data feminism, and civic data.

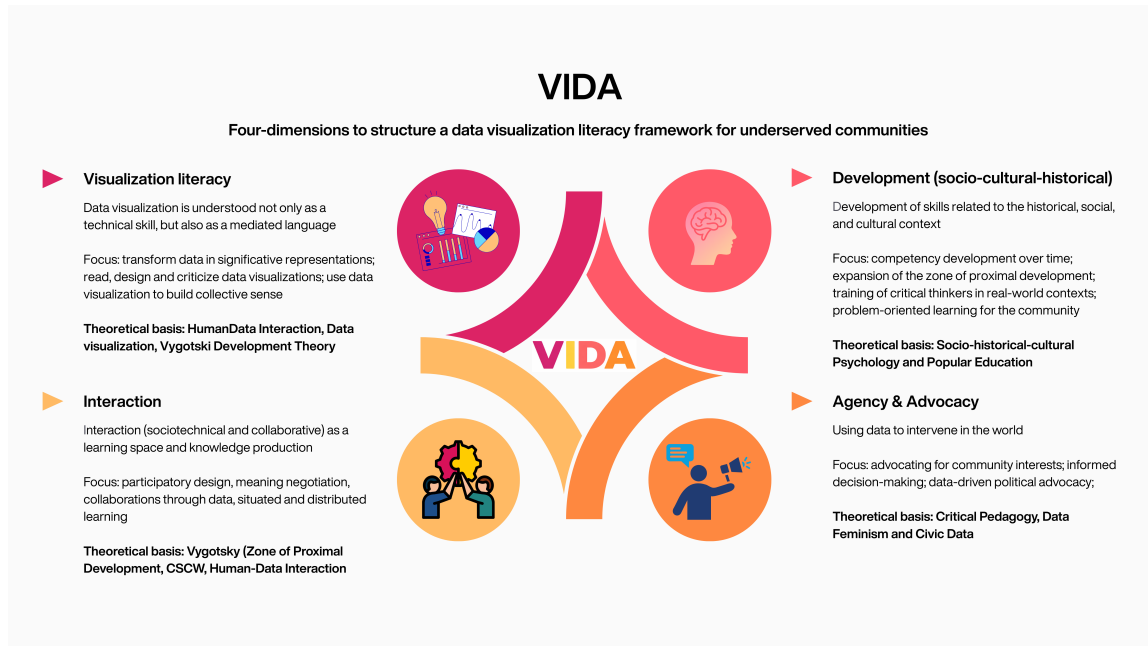


Fig. 2. Initial proposal for the VIDA framework.

4 Ethical considerations and Acknowledges

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